Data Engineer - Technical Assessment SnoreMD

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Part 1: Data Architecture & Pipeline Design (Conceptual)

Since the SnoreMD has data split across AWS (CRM + clinician notes) and Azure (sleep study exports + billing files). The objective is a unified analytics layer without forcing a full cloud migration, while maintaining PHI security and Canadian data residency.

1) Architecture Explanation:

- **ELT over ETL**: Load raw data first, then transform in the warehouse using **dbt**.
- **Snowflake as the cross-cloud integration point**: AWS and Azure land data into Snowflake independently, no direct AWS and Azure traffic required.
- **Medallion architecture**:
  - **Bronze (RAW)**: immutable raw source data + audit logs
  - **Silver (STAGING)**: cleaned, standardized, deduplicated
  - **Gold (ANALYTICS)**: star schema + marts for reporting

Source Ingestion Strategy (Batch vs Micro-Batch)

- **PostgreSQL CRM (AWS RDS)**: CDC via AWS DMS → S3 landing
  Captures row-level changes without full extracts, reducing load on production DB.
- **Scheduling REST API**: Micro-batch every 15 minutes via EventBridge → Lambda → S3
  No webhooks; 15-minute latency is acceptable for appointments operational reporting.
- **Clinician notes (AWS S3 JSON)**: Event-driven via S3 event → Glue job
  Converts/normalizes complex JSON and writes curated files for loading.
- **Sleep study exports (Azure Blob CSV)**: Event-driven batch via ADF trigger → ADLS Gen2 → Snowpipe/COPY INTO
  Device files arrive daily; trigger-on-arrival avoids polling.
- **Billing Excel (monthly)**: Blob Storage → Azure Data Factory (ADF) scheduled monthly → ADLS Gen2 → Snowflake

Data Loading & Transformations

- **Landing zones** in cloud-native storage (S3 + ADLS) retain original formats for traceability.
- **Snowflake ingestion** via **Snowpipe / COPY INTO** from both clouds.

- **dbt-core** performs transformations in Snowflake:
    - Bronze → Silver: type casting, standardization, deduplication, basic data quality gates
    - Silver → Gold: dimensional modeling (dimensions, facts), plus pre-aggregated marts

Gold Layer Modeling (Analytics)

Gold uses a **star schema** to support clinic- and time-based reporting:

- **Dimensions**: dim_patients, dim_clinics, dim_clinicians, dim_date
- **Facts**: fct_appointments, fct_sleep_studies, fct_billing
- **Mart**: mart_clinic_monthly_metrics (pre-aggregated KPIs for fast dashboard queries)

Orchestration & Operations

A master workflow runs in **AWS MWAA (Airflow)**:

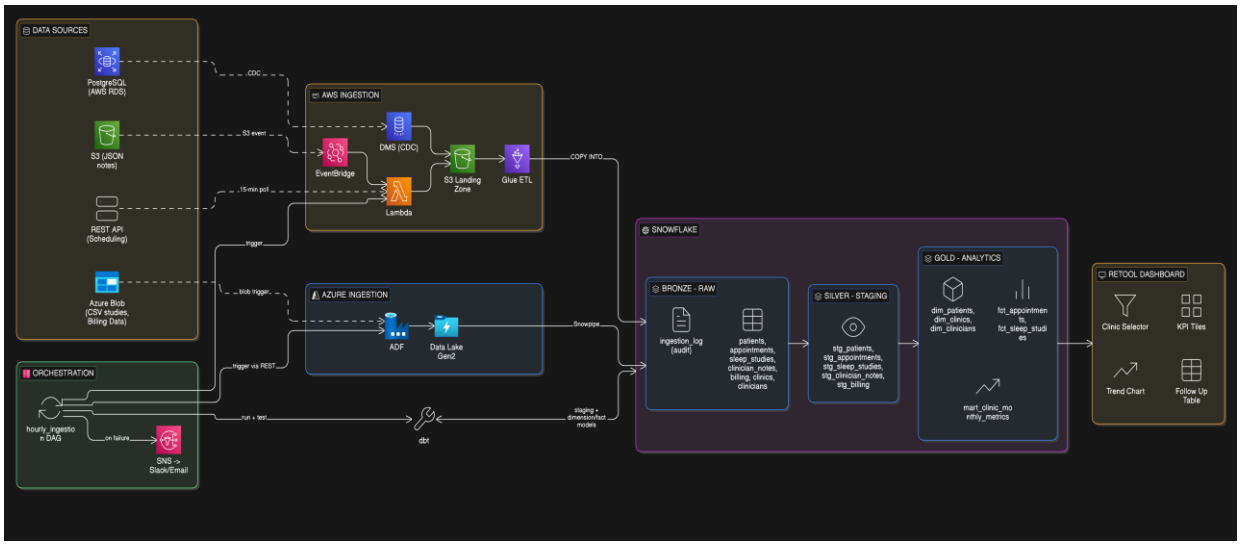
1. Run AWS ingestions in parallel (Lambda API poll + Glue notes processing + validate DMS CDC delivery)
2. Trigger **ADF pipeline** (Azure side) via REST API
3. Confirm Snowflake loads completed (Snowpipe/COPY history checks)
4. Execute dbt deps → dbt run → dbt test
5. Alert on failures via **SNS → Slack/email**

Reporting

Retool connects to Snowflake Gold/marts for:

- clinic selector
- KPI tiles (appointments, follow-ups, sleep study completion, etc.)
- trend charts + operational follow-up tables
  Pre-aggregated marts keep dashboards fast and inexpensive.

2) Architecture Diagram:



3) Services Used (AWS + Azure):

AWS

- **RDS (PostgreSQL)** - CRM source
- **DMS** - CDC from RDS to S3 landing
- **S3** - landing/processed/archive zones
- **Lambda** - REST API polling + lightweight normalization
- **EventBridge Scheduler** - 15-min trigger for API polling
- **Glue + Glue Data Catalog** - JSON preprocessing + schema discovery
- **MWAA (Airflow)** - orchestration across AWS + Azure + Snowflake
- **KMS** - key management
- **CloudWatch** - metrics/logs/alerts
- **CloudTrail** - audit logs
- **SNS** - notifications (Slack/email)

Azure

- **Blob Storage** - device CSV drop zone
- **ADLS Gen2** - raw/processed zones for Azure-sourced data
- **Data Factory (ADF)** - ingestion pipelines to Snowflake
- **Managed Identities + Azure AD** - secure auth for ADF
- **Azure Monitor** - pipeline observability
- **Private Link** - private connectivity to Snowflake

Cross-Cloud / Analytics

- **Snowflake (Canada Central)** - unified warehouse (Bronze/Silver/Gold)
- **dbt-core** - transformations + testing
- **Retool** - internal dashboard

4) Key Assumptions

1. **PHI must remain in Canada** → Snowflake deployed in **Canada Central**; no other region processing.
2. Scheduling API is **pull-based** (no webhooks) → 15-min micro-batch is acceptable.
3. Sleep study devices deliver **daily files** (not streaming) →event-triggered loads + dbt dedup by study_id.
4. Retool is **internal-only** → access scoped by clinic (row-level security).

5) Tradeoffs & Reasoning

Decision	Alternative	Why This Choice
Snowflake as hub	Redshift (AWS-only), Synapse (Azure-only)	Snowflake supports cross-cloud ingestion cleanly and enforces Canadian residency in one place.
ELT + dbt	Heavy ETL in Spark/Glue	Keeps raw data immutable, transformations version-controlled + testable, and rebuildable without re-ingest
CDC via DMS for CRM	Nightly full extracts	Reduces load on RDS, captures only changes, supports fresher reporting.
Micro-batch API polling	Streaming (Kinesis/Kafka)	Sub-minute latency isn't required; micro-batch reduces complexity and cost.
Event-driven file ingestion	Fixed interval polling	Triggers on file arrival avoids unnecessary runs and improves timeliness.
Star schema + marts	One wide denormalized table	Scales better for new dimensions and avoids expensive runtime aggregations for dashboards
MWAA orchestration	Step Functions	Airflow provides mature dependency management and easier orchestration across AWS + Azure + dbt.